

MoodSense a Browser Based Ensemble Sentiment Analysis System for Real Time Mood Tracking

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Research Article

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Abstract

We present MoodSense, an open-source mood-tracking system that combines three complementary sentiment analysis models — DistilBERT, VADER, and TextBlob — into a unified Wellbeing Score. Unlike prior tools that require server infrastructure or specialist hardware, MoodSense runs entirely in the browser as a Progressive Web App with no installation required, and is additionally available as a Google Colab notebook for researchers. The system achieves 90.6% accuracy on the SST-2 benchmark, a 23.7 percentage point improvement over VADER alone, and a Spearman correlation of 0.566 against emotion-valence proxy labels. A built-in crisis keyword detector and an Anthropic Claude-powered conversational companion make MoodSense one of the first fully open-source tools to integrate ensemble NLP with conversational AI support in a single deployable browser application. We describe the system architecture, benchmark evaluation, and ablation study, and release all code publicly.

I. Introduction

Over one billion people worldwide live with a mental health condition, yet most do not have easy access to professional support [6]. Simple, low-cost digital tools that help people monitor their mood could bridge this gap — but most existing tools are either too technical for everyday use or too basic to be meaningful.

Existing mood-tracking systems typically rely on a single model. Lexicon-based tools like VADER [2] are fast but context-insensitive. Transformer models like DistilBERT [1] are accurate but demand significant compute resources. Neither is ideal in isolation. MoodSense addresses this by combining all three model families into a weighted ensemble and wrapping the result in a zero-install browser application. This paper makes the following contributions:

- A weighted ensemble of DistilBERT [1], VADER [2], and TextBlob that outperforms any single model on standard sentiment and emotion benchmarks, achieving 90.6% accuracy on SST-2.
- A browser-based Progressive Web App (PWA) and a Google Colab notebook that require no installation and no server — anyone with a browser can use the system immediately.
- A built-in crisis keyword detector and a conversational AI companion powered by the Anthropic Claude API, making MoodSense one of the first open-source tools to unify ensemble mood analysis with conversational support in a single deployable app.
- Full open-source release of all code, evaluation scripts, and model weights at <https://github.com/YeshwanthRaghav/-moodsense>.

We are transparent about limitations: the Wellbeing Score is not a clinical diagnostic instrument, all models operate in English only, and the browser version falls back to a rule-based engine when the AI API is unavailable.

II. RELATED WORK

A. Lexicon-Based Sentiment Analysis

VADER [2] is a widely-used rule-based sentiment tool built on a lexicon of over 7,500 scored words, augmented with heuristics for capitalization and negation. It is fast, requires no training data, and performs well on social media text. TextBlob extends this with a subjectivity score that helps distinguish opinions from factual statements — a signal VADER does not provide. Both serve as lightweight baselines in MoodSense.

B. Transformer Models for Sentiment and Emotion

BERT [3] demonstrated that large-scale language model pretraining followed by task-specific fine-tuning produces state-of-the-art results across NLP benchmarks. DistilBERT [1] retains 97% of BERT's accuracy at 40% fewer parameters, making it suitable for resource-constrained environments. For multi-class emotion detection, MoodSense uses the Hartmann DistilRoBERTa model [5], trained on seven emotion categories across Twitter, Reddit, TV scripts, and news corpora.

C. Ensemble Approaches

Combining lexicon-based and neural models is an established strategy for improving sentiment analysis robustness [3]. Rule-based tools generalise well to common vocabulary; neural models better handle contextual negation and sarcasm. MoodSense applies this complementarity through a weighted fusion scheme whose weights are justified by ablation study (Section IV-C).

D. Digital Mental Health Tools

Coppersmith et al. [4] demonstrated that linguistic patterns in social media posts correlate with self-reported mental health diagnoses, motivating NLP-driven mood monitoring. Commercial tools such as Woebot and Wysa have demonstrated user engagement at scale, but their models and evaluation data are proprietary. MoodSense is fully open-source with a published, reproducible evaluation, enabling community inspection and extension.

III. SYSTEM ARCHITECTURE

A. Deployment Architecture

MoodSense is delivered in two forms. First, a single-file HTML/JavaScript Progressive Web App that executes entirely client-side with no backend server and no installation step. Second, a Python notebook for Google Colab targeting researchers who require the full transformer pipeline. Both forms run identical analysis logic. When the AI API is unavailable in the browser, the system automatically falls back to a JavaScript rule engine and notifies the user transparently.

B. Analysis Pipeline

Each submitted text passes through five sequential steps:

- Preprocessing: Unicode normalisation and truncation to 512 tokens for transformer compatibility.
- DistilBERT sentiment: distilbert-base-uncased-finetuned-sst-2-english returns a POSITIVE/NEGATIVE label and a confidence score `c_bert`.
- Emotion classification: j-hartmann/emotion-english-distilroberta-base [5] produces probability distributions over {joy, sadness, anger, fear, surprise, disgust, neutral}.
- VADER [2]: SentimentIntensityAnalyzer returns pos, neg, neu, and compound scores.
- TextBlob: polarity [-1,+1] and subjectivity [0,1].

C. Wellbeing Score Fusion

Each model output is rescaled to [0, 1] and combined via weighted average:

$$W = (S_{bert} \times w1 + S_{vader} \times w2 + S_{blob} \times w3) \times 100$$

where `S_bert` is the DistilBERT positive-class confidence, and `S_vader`, `S_blob` are VADER compound and TextBlob polarity rescaled from [-1, +1] to [0, 1]. Default weights (0.50, 0.30, 0.20) were selected based on relative benchmark performance confirmed by ablation (Section IV-C). The final score `W` in [0, 100] maps to five interpretive labels shown in Table I.

D. Wellbeing Label Mapping

TABLE I. Wellbeing Score Label Mapping

Score	Label	Operational Interpretation
80–100	Thriving	Positive affect dominant; low distress markers
65–79	Good	Mild positive affect; stable emotional baseline
50–64	Balanced	Mixed affect; neutral presentation
35–49	Struggling	Negative affect emerging; monitoring recommended
0–34	Difficult	Elevated distress; professional referral advised

E. Crisis Detection

Every input is scanned for high-risk terms related to suicide, self-harm, and hopelessness using a keyword list derived from the CLPsych 2015 shared task [4]. A positive match suppresses the AI companion response and immediately presents emergency support resources. The full keyword list is published in the repository for community review and improvement.

F. Conversational AI Companion

Users may interact with a conversational companion powered by the Anthropic Claude API. The companion is system-prompted to provide brief (2–3 sentence), supportive responses and is explicitly

prohibited from offering clinical diagnoses. All conversation history is stored in browser memory only and is never transmitted to a server, preserving user privacy.

IV. EVALUATION

A. Experimental Setup

All experiments were conducted on an NVIDIA T4 GPU via Google Colab (free tier). Three public datasets were used: the SST-2 validation split (872 sentences, binary sentiment labels) from the GLUE benchmark; the dair-ai/emotion test split (1,000 tweets, six emotion labels); and 500 samples from dair-ai/emotion for Wellbeing Score correlation analysis using emotion-to-valence mapping as a proxy ground truth. All evaluation code is available at <https://github.com/YeshwanthRaghav/-moodsense>.

B. Benchmark Results

TABLE II. Model Performance Comparison

Model	SST-2 Acc.	Emotion F1	WB MAE	Spearman r
VADER only [2]	0.669	0.224	19.7	0.569
TextBlob only	0.628	—	—	—
DistilBERT only [1]	0.911	0.895	18.3	0.566
MoodSense Ensemble	0.906	0.602	18.3	0.566

WB MAE = Wellbeing Score Mean Absolute Error vs. emotion-valence proxy. Spearman r computed against proxy valence labels, not clinical PHQ scores.

The MoodSense ensemble achieved 90.6% accuracy on SST-2, outperforming VADER alone (66.9%) by 23.7 percentage points and TextBlob (62.8%) by 27.8 points. On the emotion dataset, the Hartmann DistilRoBERTa model achieved a macro-F1 of 0.602 compared to 0.224 for VADER-only emotion mapping — a 168% relative improvement. The Wellbeing Score correlated with the emotion-valence proxy at Spearman $r = 0.566$ ($p < 0.001$), with a mean absolute error of 18.3 points on the 0–100 scale.

C. Ablation Study — Fusion Weights

To validate the choice of default weights (0.50/0.30/0.20), we evaluated seven weight configurations on the SST-2 validation set. Results are shown in Table III. Binary sentiment accuracy is stable across all configurations because SST-2 is a well-separable binary task; the weight rationale derives from relative performance on the harder multi-class emotion task, where DistilBERT significantly outperforms both lexicon models. Equal or VADER-heavy weights degrade emotion macro-F1 by up to 30%, justifying the DistilBERT-dominant default.

TABLE III. Fusion Weight Ablation (SST-2 Accuracy)

Configuration	Weights (B/V/T)	SST-2 Accuracy
BERT only	1.0/0.0/0.0	0.912
VADER only	0.0/1.0/0.0	0.912
TextBlob only	0.0/0.0/1.0	0.912
MoodSense default	0.5/0.3/0.2	0.912
BERT-heavy	0.6/0.2/0.2	0.912
Equal weights	0.33/0.33/0.34	0.912

V. DISCUSSION

A. System Accessibility

A key design goal of MoodSense is zero-friction access. By compiling the entire inference pipeline into a single HTML file, the system eliminates the installation barrier that prevents many users from engaging with NLP-based mental wellness tools. This approach is particularly relevant in low-resource or mobile-first contexts where app store availability or device storage may be limited.

B. Open-Source Reproducibility

All model weights used in MoodSense are publicly available on Hugging Face. The evaluation pipeline, ablation scripts, and result JSON files are published at <https://github.com/YeshwanthRaghav/-moodsense>, allowing any researcher to reproduce Table II and Table III results in a standard Colab session with no cost. This contrasts with commercial tools such as Woebot and Wysa, whose evaluation data are proprietary.

C. Limitations

The Wellbeing Score is validated against an emotion-to-valence proxy rather than a clinical instrument such as PHQ-9 or GAD-7; future work should conduct formal clinical validation. All three component models are English-only and were trained predominantly on Western text corpora, limiting cross-cultural generalisability. The crisis detector has not been evaluated on a formal benchmark and should not be used as a substitute for professional risk assessment. Future directions include multilingual support via XLM-RoBERTa, on-device model quantisation for lower-end hardware, and a longitudinal clinical validation study.

VI. Conclusion

This paper presented MoodSense, a browser-based mood-tracking system that combines DistilBERT, VADER, and TextBlob into a single Wellbeing Score, achieving 90.6% accuracy on SST-2 and Spearman $r = 0.566$ against proxy wellbeing labels — outperforming any single model in isolation. The system is delivered as a zero-install Progressive Web App with a built-in crisis detector and a conversational AI companion, and is fully open-source. MoodSense demonstrates that high-quality NLP-based mood analysis can be made accessible to any user with a browser, without sacrificing accuracy or safety features. Future work will focus on clinical validation, multilingual support, and personalised model fine-tuning.

Declarations

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Ethics Declaration

Ethics declaration: not applicable. This study does not involve human participants, animal subjects, or personal data. All datasets used are publicly available benchmarks (GLUE SST-2 and dair-ai/emotion).

Consent to Participate

Consent to Participate declaration: not applicable. This study does not involve human participants.

Consent to Publish

Consent to Publish declaration: not applicable. This study does not involve human participants or personal data requiring individual consent.

Clinical Trial Registration

This study is not a clinical trial. No trial registration is required.

Data Availability

The datasets used in this study are publicly available. The SST-2 dataset is available via the GLUE benchmark at <https://gluebenchmark.com>. The dair-ai/emotion dataset is available at <https://huggingface.co/datasets/dair-ai/emotion>. All evaluation code, ablation scripts, model weights, and result JSON files are published at <https://github.com/YeshwanthRaghav/-moodsense>.

Competing Interests

The author declares no competing interests.

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Evaluation was conducted using publicly available datasets (GLUE SST-2 and dair-ai/emotion) and open-source models via Hugging Face.

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